

*You MUST lay out your work clearly and concisely and, in this topic more than ever, must not cut corners. Everything must be explained clearly and you must be clear in your conclusions.*

1. The manufacturers of a popular brand of crisp are running a promotion in which some of their bags of crisps contain a voucher for a prize. These prizes range from small prizes, such as a free bag of crisps, to larger prizes, like iPods, Blu-Ray players, big-screen televisions, and even a holiday to the United States. The manufacturer claims that, overall, one in five packets of crisps contains a prize. Peter believes, however, that this is not the case, and sets out to test this hypothesis by purchasing 25 packs of crisps.

- (a) Write down Peter's null and alternative hypothesis.
- (b) What is the critical region for Peter's test, at the 10% level of significance?

Peter opens his bags of crisps, and finds that only two of them contains a prize.

- (c) What is the conclusion of the hypothesis test?